Class Activity: "Spot the Essential Features"

Step 1: Set the Scene

Imagine you are training an AI to recognise birds. Normally, it sees birds sitting on trees with the sky in the background. But in the real world, birds can appear anywhere, even against weird, irrelevant backgrounds like fractals, buildings, or graffiti. How can the AI learn to focus on what really matters?

Step 2: Group Task
Each group gets to:

1. Sketch a simple bird
2. Around the bird, add 3 different backgrounds — e.g.:
    - Trees/sky (natural).
    - Random shapes/patterns (like zig-zags or spirals).
    - A totally unusual one (e.g., buildings, boxes, or even doodles).
3. Discuss as a group:
    - Which features of the bird stay constant across all three drawings?
    - Which parts are just background "noise"?
    - If you were designing an AI, what features would you make it focus on?

Step 3: Share & Debrief

- Each group explains what they chose as the essential bird features (wings, beak, body shape).
- Connect it to **PixMix**: mixing irrelevant or artificial backgrounds helps AI ignore noise and focus on robust features.

Note: You can use GenAI to generate these images.